

# PMx Weekly Digest: July 26, 2026

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## Paper 1: From Radiocopper to Cold Copper: Mechanistic Modeling and Simulation to Define C

### Bibliographic Information

- **Title:** From Radiocopper to Cold Copper: Mechanistic Modeling and Simulation to Define Clinical Response Criteria and Biomarkers for VTX-801 in Wilson Disease
- **Authors:** Lindauer A, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026
- **DOI:** 10.1002/psp4.70153

**Paper Type:** POPK\_PKPD

### Quick Take

This paper presents a mechanistic PBPK model of copper metabolism that was used to define clinical response criteria and biomarkers for VTX-801, an AAV gene therapy for Wilson disease. The model integrates radiocopper kinetics, disease pathophysiology, and preclinical gene therapy data to establish a 48/2-h plasma radioactivity ratio cut-off of 0.4 for identifying responders in a Phase 1/2 trial, and identifies relative exchangeable copper (REC) as a promising non-radioactive biomarker. This work is essential reading for pharmacometricians involved in rare disease drug development, gene therapy, or quantitative systems pharmacology.

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### Executive Summary

Lindauer et al. developed a mechanistic, minimal PBPK model of human copper metabolism to support biomarker qualification and dose-response assessment for VTX-801, an AAV-based gene therapy for Wilson disease (WD). The model explicitly distinguishes between ceruloplasmin-bound and non-ceruloplasmin-bound copper (NCC) and incorporates ATP7B-dependent biliary excretion and ceruloplasmin loading. It was calibrated using literature data from healthy subjects, heterozygous carriers, and WD patients, as well as clinical radiocopper data from a dedicated pilot study (COMET). Preclinical dose-response data from WD mice treated with VTX-801 were used to derive activity factors (AF) corresponding to restored ATP7B function, with 15% and 40% selected as minimal and optimal efficacy targets. Simulations demonstrated that the 48/2-h plasma radioactivity ratio can effectively differentiate responders from non-responders, leading to a pre-specified cut-off of 0.4 for the ongoing clinical trial. Cold copper simulations further identified relative exchangeable copper (REC) as a sensitive, non-radioactive biomarker for treatment response. This work provides a robust quantitative framework for assessing copper-related biomarkers and guiding gene therapy development in WD.

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## Scientific Context & Motivation

Wilson disease (WD) is a rare autosomal recessive disorder caused by mutations in the ATP7B copper transporter, leading to impaired biliary copper excretion and deficient ceruloplasmin loading, resulting in toxic copper accumulation. Current standard-of-care (chelators, zinc) requires lifelong therapy and monitoring, creating a need for disease-modifying therapies like VTX-801, an AAV-based gene therapy. A critical challenge is how to measure treatment response in the presence of background therapy that can normalize traditional biomarkers (e.g., plasma copper, urinary copper). The radiocopper balance test, particularly the 48/2-h plasma radioactivity ratio, has been shown to discriminate WD patients from healthy subjects with high sensitivity and specificity, but its use as a responder criterion for gene therapy had not been established. Previous copper kinetic models (Scott, Buckley, Harvey) described physiological kinetics but did not capture WD pathophysiology or support biomarker-based assessment of gene therapy. This work fills that gap by developing a model that explicitly incorporates ATP7B-dependent processes and links preclinical dose-response data to human biomarker predictions.

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## ✂ Methodological Snapshot

The authors developed a minimal PBPK model of human copper metabolism in NONMEM, comprising compartments for plasma (NCC and Cu-CP), liver (extracellular, precursor, storage), tissue, deep tissue, GI tract, and urine. The model was calibrated using literature data from 18 published studies (1954 onwards) and clinical data from the COMET study (3 healthy, 3 heterozygotes). ATP7B dysfunction in WD was modeled via separate activity factors for ceruloplasmin binding ( $AF_{cp}$ ) and biliary excretion ( $AF_{bil}$ ), with heterozygote estimates of 0.369 and 0.101, respectively. Preclinical dose-response data from WD mice treated with VTX-801 were fit with a sigmoid  $E_{max}$  model to derive  $AF$  values of 0.15 (minimal) and 0.4 (optimal) for human efficacy targets. Simulations of 42,000 virtual patients (2000 per  $AF$  from 0 to 1) were used to define the 48/2-h ratio cut-off of 0.4 for responder identification. Cold copper simulations at steady-state (1.4 mg/day oral intake) identified REC as the most sensitive non-radioactive biomarker.

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## ☒ Structural Model Breakdown

The model is a minimal PBPK model with the following compartments: (1) Plasma NCC (non-ceruloplasmin-bound copper), (2) Plasma Cu-CP (ceruloplasmin-bound copper), (3) Liver extracellular space, (4) Liver precursor pool, (5) Liver storage pool, (6) Tissue (peripheral), (7) Deep tissue, (8) Absorption compartment (duodenum/stomach), (9) Distal gut, (10) Feces, (11) Urine. Copper enters the system via oral absorption ( $k_{abs_{liv}}$ ,  $k_{abs_{gutD}}$ ) or IV injection. NCC distributes between plasma, liver, tissue, and deep tissue via first-order rate constants. In the liver, NCC enters a precursor pool where it can be: (a) incorporated into ceruloplasmin ( $k_{livPre_{plCP}}$ , ATP7B-dependent, modulated by  $AF_{cp}$ ), (b) excreted into bile ( $k_{livPre_{out}}$ , ATP7B-dependent, modulated by  $AF_{bil}$ ), or (c) stored in the liver storage pool ( $k_{livPre_{livStore}}$ ,  $k_{livStore_{livPre}}$ ). Biliary copper enters the GI tract where 25% is reabsorbed and 75% is eliminated in feces. Cu-CP distributes between plasma and tissue with fixed rate constants based on ceruloplasmin kinetics and is eliminated with a half-life of 5.5 days. NCC is also eliminated via urine ( $k_{pl_{urine}}$ ) and sweat

( $k_{tis\_sweat}$ ). Chelator therapy increases renal elimination ( $Chel_{eff} = 16.8$ -fold). Zinc reduces absorption (40.1%) and increases plasma-tissue distribution (1.2-fold). Disease-specific alterations in WD include reduced  $k_{liv\_livPre}$  (15.3% decrease) and increased tissue-to-plasma distribution rates (2.43-fold for  $k_{tis\_pl}$ , 27.6-fold for  $k_{tis2\_pl}$ ).

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## Detailed Methodological Analysis

### Modeling Approach

Minimal physiologically based pharmacokinetic (PBPK) model with compartments for plasma (NCC and Cu-CP), liver (extracellular, precursor, storage), tissue, deep tissue, GI tract (absorption, distal gut), and urine. The model explicitly distinguishes between ceruloplasmin-bound copper (Cu-CP) and non-ceruloplasmin-bound copper (NCC). ATP7B-dependent processes are modeled via activity factors ( $AF_{cp}$  for ceruloplasmin binding,  $AF_{bil}$  for biliary excretion). Biliary copper reabsorption (25%) is implemented by splitting the biliary elimination rate. The model was developed in NONMEM 7.4.3. Oral absorption parameters were fixed from an earlier model version. Chelator and zinc effects were modeled as fractional changes on relevant rate constants.

### Data Sources

Literature data from 18 published human studies (1954 onwards) including healthy individuals, heterozygous carriers, and WD patients, with radiocopper ( $^{64}Cu$ ,  $^{67}Cu$ ,  $^{65}Cu$ ) administered orally or intravenously. Data included plasma, fecal, urinary, and liver radioactivity (PET and scintillation). Clinical data from the COMET study (EudraCT: 2019-001157-13): 3 healthy adults and 3 WD carriers receiving a single IV dose of [ $^{64}Cu$ ]CuCl<sub>2</sub>, with 84 plasma samples up to 96 h post-dose (16 below LOD excluded). External validation data from Czlonkowska et al. provided 48/2-h and 24/2-h serum ratios for 86 WD patients, 40 carriers, and 18 healthy controls. Preclinical data from two mouse studies (WT and WD mice, n=6/group) with VTX-801 doses of 0, 1.5, 5, and  $15 \times 10^{12}$  vg/kg, measuring ceruloplasmin oxidase activity and  $^{64}Cu$  fecal excretion.

### Estimation Methods

NONMEM version 7.4.3 was used with the first-order (FO) estimation method. Parameters were estimated on the log-scale and back-transformed by exponentiation. Inter-individual variability (IIV) was not estimated due to the small sample size; instead, a fixed 15% CV was assumed for all rate constants in simulations. Residual variability was modeled as proportional error, with separate estimates for each data type (plasma literature: 0.698, plasma COMET: 0.176, feces: 0.711, urine: 0.611, liver: 0.203, CP-bound: 0.441).

### Model Evaluation

Goodness-of-fit plots and residual plots (Figures S1, S2) showed good agreement. Visual predictive checks (VPCs) for COMET data demonstrated that the 5th-95th percentile prediction intervals (from 2000 simulated subjects per population) adequately captured the observed data. External validation against the independent Czlonkowska et al. dataset showed excellent agreement for heterozygotes and healthy subjects, with slight overprediction for WD patients that remained well below the diagnostic cut-off. An alternative VPC including parameter uncertainty was provided

in Figure S7. A high-variability scenario (30% IIV/20% RUV) was also simulated and compared (Figure S6).

### Covariate Analysis

Binary covariates were used to capture disease-specific differences in WD patients: a 15.3% decrease in the rate from liver extracellular space to precursor pool ( $k_{liv\_livPre}$ ), a 2.43-fold increase in tissue-to-plasma distribution ( $k_{tis\_pl}$ ), and a 27.6-fold increase in deep tissue-to-plasma distribution ( $k_{tis2\_pl}$ ). These were attributed to saturation of binding sites and excess unbound copper in WD. A COMET-specific covariate reduced the gut transfer rate ( $k_{gutD\_feces}$ ) by 60.7% to account for slower fecal elimination in that study. Chelator therapy was modeled as a 16.8-fold increase in renal elimination. Zinc therapy was modeled as a 40.1% reduction in absorption and a 1.2-fold increase in plasma-tissue distribution. No formal covariate selection procedure (e.g., stepwise forward/backward) was reported; covariates were pre-specified based on physiological plausibility.

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### Statistical Rigor Assessment

The model was developed using a combination of literature data (18 studies) and clinical data (COMET study,  $N = 6$ ), with external validation against an independent dataset ( $N = 144$ ). The use of the FO estimation method in NONMEM is appropriate given the sparse individual data, but the inability to estimate IIV is a limitation. The authors addressed this by assuming 15% CV on all rate constants and testing a higher variability scenario (30% IIV/20% RUV) to assess robustness. The model comparison for single vs. dual AF was supported by a  $\Delta OFV$  of 40.657, which is highly significant ( $p < 0.001$  for 1  $df$ ). Parameter precision was generally good, with RSE values  $< 30\%$  for most structural parameters, though some covariate effects (e.g., WD effect on  $k_{liv\_livPre}$ , RSE 64.8%) were less precisely estimated. The external validation against the Czlonkowska dataset is a strength, though the overprediction for WD patients suggests the model may be slightly conservative for this population. The use of 2000 virtual subjects per scenario for simulations provides adequate precision for the decision criteria. No formal sensitivity analysis was reported, but the exploration of high-variability scenarios partially addresses this.

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### Key Findings

The model successfully captures the distinct plasma, fecal, and urinary radiocopper kinetics in healthy subjects, heterozygous carriers, and WD patients, with separate activity factors for ceruloplasmin binding ( $AF_{cp}$ ) and biliary excretion ( $AF_{bil}$ ) providing a significantly better fit than a single  $AF$  ( $\Delta OFV = 40.7$ ). In heterozygotes,  $AF_{cp}$  was estimated at 0.369 (RSE 5.5%) and  $AF_{bil}$  at 0.101 (RSE 24.8%), indicating that biliary excretion is more severely impaired at intermediate ATP7B function. Preclinical dose-response data in WD mice identified 15% and 40% of normal ATP7B activity as minimal and optimal efficacy targets, respectively. Simulations established a 48/2-h plasma radioactivity ratio cut-off of 0.4 for identifying VTX-801 responders, providing  $>80\%$  probability of correct classification. Cold copper simulations identified relative exchangeable copper (REC) as the most sensitive non-radioactive biomarker, capable of detecting changes

at  $AF \geq 0.05$ , while 24-h urinary copper required  $AF \geq 0.2$  to show meaningful change. REC was also unaffected by background chelator or zinc therapy, unlike urinary copper. The model was externally validated against an independent dataset of 48/2-h and 24/2-h ratios, showing excellent agreement for heterozygotes and healthy subjects.

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## 🔗 Clinical & Regulatory Implications

The model directly supported the design of a Phase 1/2 gene therapy trial for VTX-801 by defining a 48/2-h plasma radioactivity ratio cut-off of 0.4 to identify responders and guide safe withdrawal of standard-of-care therapy. This criterion was chosen to ensure >80% probability of correct classification across simulated scenarios. The model also identified relative exchangeable copper (REC) as a promising non-radioactive biomarker that is sensitive to small changes in ATP7B activity ( $AF \geq 0.05$ ) and unaffected by background chelator or zinc therapy, making it suitable for broader clinical use. In contrast, 24-h urinary copper was less sensitive and confounded by background treatments. The simulations suggest that restoring 15–40% of normal ATP7B function may be sufficient for clinical benefit, consistent with the asymptomatic status of heterozygous carriers.

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## Strengths & Limitations

### Strengths

- Comprehensive integration of data from 70+ years of copper kinetic studies, including literature, clinical, and preclinical sources.
- Mechanistic distinction between ceruloplasmin-bound and non-ceruloplasmin-bound copper, with separate activity factors for the two ATP7B-dependent processes.
- Successful external validation against an independent dataset of 144 subjects.
- Direct translation of model simulations into a clinically actionable decision criterion (48/2-h ratio cut-off of 0.4) for a first-in-human gene therapy trial.
- Identification of REC as a sensitive, non-radioactive biomarker that is unaffected by background therapy, addressing a key practical limitation.
- Provision of NONMEM model code and campsis simulation code in the Supporting Information, enhancing reproducibility.
- Clear and well-structured presentation of the model development, validation, and application.

### Limitations (Acknowledged by Authors)

- Small sample size ( $N = 6$ ) in COMET study precluded estimation of IIV; fixed 15% CV was assumed.
- Liver copper predictions ( $\sim 75 \mu\text{g/g}$ ) in untreated WD patients were lower than typical clinical values ( $> 250 \mu\text{g/g}$ ), suggesting underprediction.
- Fecal data showed unexplained  $\sim 60\%$  slower gut transit in COMET compared to literature, increasing uncertainty in fecal-based criteria.
- Limited availability of published fecal data hindered robust model calibration.

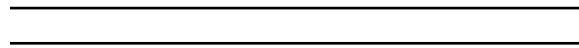
- The model lacks nonlinear tissue binding; altered tissue-plasma kinetics in WD were captured via binary covariates, which may not fully capture dynamics during treatment.
- Different activity factors for  $AF_{cp}$  and  $AF_{bil}$  in heterozygotes (0.37 vs. 0.10) require further investigation.
- Daily copper intake was fixed at 1.4 mg, ignoring geographical and dietary variability.
- Chelator effect estimate (16.8-fold increase) relied heavily on a single study.
- The model could not predict efficacious gene therapy dose in humans due to species differences in vector kinetics and immunology.

### Limitations (Expert Review)

- The use of the FO estimation method, while pragmatic, may introduce bias in parameter estimates, especially for models with significant nonlinearity.
- The assumption that IIV is the same (15% CV) for all rate constants is physiologically implausible; some parameters (e.g., absorption, biliary excretion) are likely more variable than others.
- The model does not account for the potential impact of gene therapy on the immune system or vector-related toxicity, which could affect biomarker interpretation.
- The external validation dataset (Czlonkowska) used serum rather than plasma, which may introduce a systematic bias, though the authors note good agreement.
- The model's underprediction of liver copper in WD patients suggests that the liver sub-model may need revision, potentially by incorporating a more detailed representation of hepatic copper storage and turnover.
- The power function linking  $AF_{cp}$  and  $AF_{bil}$  for simulations under treatment is empirical and not mechanistically justified.
- No formal model selection criteria (e.g., AIC, BIC) were reported for the structural model development.

### Generalizability

The model is specific to copper metabolism and Wilson disease, but the framework for linking preclinical gene therapy dose-response to human biomarker predictions is broadly applicable to other monogenic disorders treated with gene therapy. The approach of using a mechanistic model to define responder criteria for a first-in-human trial is a paradigm that could be adopted for other rare diseases.



### Figures & Tables

- **Figure 1:** Schematic of the updated copper model showing compartments for plasma (NCC and Cu-CP), liver (extracellular, precursor, storage), tissue, deep tissue, GI tract, and urine, with ATP7B-dependent processes ( $AF_{cp}$ ,  $AF_{bil}$ ) highlighted.
  - *Significance:* Provides the structural framework for the entire modeling effort, illustrating the key physiological compartments and the dual role of ATP7B in ceruloplasmin loading and biliary excretion.
- **Figure 2:** Observed and predicted radioactivity-time profiles in plasma (total and CP-bound), feces, urine, and liver for healthy, heterozygote, and WD subjects.

- *Significance*: Demonstrates the model’s ability to simultaneously capture diverse datasets across multiple matrices and populations, supporting its mechanistic validity.
- **Figure 3**: Visual predictive check (VPC) for COMET study data, showing observed plasma and fecal radioactivity profiles against 5th-95th percentile prediction intervals.
  - *Significance*: Validates the model’s predictive performance on an independent clinical dataset, confirming that the assumed 15% CV IIV and 15% CV RUV are reasonable.
- **Figure 4**: External validation comparing simulated vs. observed 48/2-h and 24/2-h plasma radioactivity ratios for healthy, heterozygote, and WD subjects from Członkowska et al.
  - *Significance*: Provides strong external validation of the model’s ability to predict the key diagnostic ratio, with excellent agreement for heterozygotes and healthy subjects, and conservative overprediction for WD patients.
- **Figure 5**: Panel A: Simulated plasma radiocopper profiles for different  $AF$  values. Panel B: 48/2-h ratio as a function of  $AF_{cp}$ . Panel C: Probability of correct classification vs. 48/2-h cut-off for optimistic ( $AF = 0.15$ ) and realistic ( $AF = 0.4$ ) targets.
  - *Significance*: Central to the clinical decision-making framework, showing how the model was used to select the 48/2-h ratio cut-off of 0.4 that balances sensitivity and specificity for identifying VTX-801 responders.
- **Figure 6**: Simulations of cold copper biomarkers (total serum copper, NCC, 24-h urinary copper, REC) across healthy, untreated WD, and WD with background therapy, as a function of  $AF$ .
  - *Significance*: Demonstrates that REC is the most sensitive non-radioactive biomarker for detecting changes in ATP7B activity, outperforming traditional markers, and is unaffected by background chelator or zinc therapy.
- **Table 1**: Final parameter estimates for the copper PBPK model, including rate constants, activity factors, covariate effects, and residual error terms.
  - *Significance*: Provides the complete parameterization of the model, including key estimates such as  $AF_{cp} = 0.369$  and  $AF_{bil} = 0.101$  for heterozygotes, and the 16.8-fold increase in renal elimination with chelator therapy.

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## Code & Reproducibility Assessment

NONMEM model code is provided in the Supporting Information. The campsis simulation code is also mentioned as available in the Supporting Information. No public repository link is provided.

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## Supplementary Materials

Supporting Information includes NONMEM model code, campsis simulation code, additional goodness-of-fit plots (Figures S1, S2), dose-response model fits for mice (Figures S3, S4, Table S2), high-variability scenario simulations (Figures S5, S6), VPC with parameter uncertainty (Figure S7), and REC response to chelator/zinc (Figure S8). An overview of the 18 literature references is provided in Table S1.

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## Future Directions

The model raises several important questions for future research. First, the differential activity factors for ceruloplasmin binding and biliary excretion in heterozygotes (0.37 vs. 0.10) warrant further investigation, potentially through dedicated mechanistic studies or analysis of individual patient data. Second, the assumption of linear tissue binding may need to be refined as data from treated WD patients become available, particularly to understand copper redistribution dynamics during recovery. Third, the model's prediction of liver copper content ( $\sim 75 \mu\text{g/g}$ ) in untreated WD patients is lower than typical clinical values ( $> 250 \mu\text{g/g}$ ), suggesting that the liver sub-model may need revision, possibly by incorporating a more detailed representation of hepatic copper storage and turnover. Fourth, the model could be extended to include other copper-related disorders or therapies with different mechanisms of action. Finally, the framework could be adapted to support model-based adaptive trial designs, where the 48/2-h ratio is used not just as a binary responder criterion but as a continuous endpoint for dose optimization.

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## Expert Commentary

This is a tour de force in mechanistic modeling applied to a rare disease with a well-understood monogenic defect. The authors have skillfully integrated data spanning over 70 years of copper kinetic studies, from early radiotracer experiments to modern preclinical gene therapy data. The key innovation is the use of a dual activity factor ( $AF_{cp}$  and  $AF_{bil}$ ) to separately capture the two ATP7B-dependent processes, which was statistically justified by a significant improvement in model fit ( $\Delta OFV = 40.7$ ). The finding that heterozygous carriers have a higher  $AF$  for ceruloplasmin binding (0.37) than for biliary excretion (0.10) is biologically plausible and suggests a differential sensitivity of these pathways to ATP7B dysfunction. From a regulatory perspective, the use of the model to pre-specify a responder criterion (48/2-h ratio  $> 0.4$ ) for a first-in-human gene therapy trial is exemplary and demonstrates the value of MIDD in rare disease drug development. The exploration of REC as a non-radioactive biomarker is particularly forward-thinking, as it addresses the practical limitations of radiocopper studies in larger trials. One limitation that warrants attention is the assumption of linear tissue binding; the model captures the altered tissue-plasma distribution in WD via binary covariates, but this may not fully capture the dynamics of copper redistribution as ATP7B function is restored. Additionally, the arbitrary assignment of 15% CV for inter-individual variability, while pragmatic, could be refined as more individual-level data become available. Overall, this work sets a high standard for model-informed development of gene therapies.

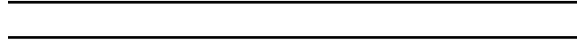
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## Bottom Line

This paper presents a comprehensive mechanistic PBPK model of human copper metabolism that successfully integrates radiocopper kinetics, disease pathophysiology, and preclinical gene therapy data to define clinically actionable biomarker thresholds for VTX-801 in Wilson disease. The model's ability to simulate both 'hot' (radioactive) and 'cold' (endogenous) copper biomarkers, and to link ATP7B activity factor to the 48/2-h plasma ratio and relative exchangeable copper (REC), provides a robust quantitative framework for trial design and responder identification. For



practicing pharmacometricians, this work exemplifies how a well-constructed mechanistic model can bridge preclinical and clinical data to directly inform regulatory-grade decision criteria in a rare disease setting.



## ☒ Figures

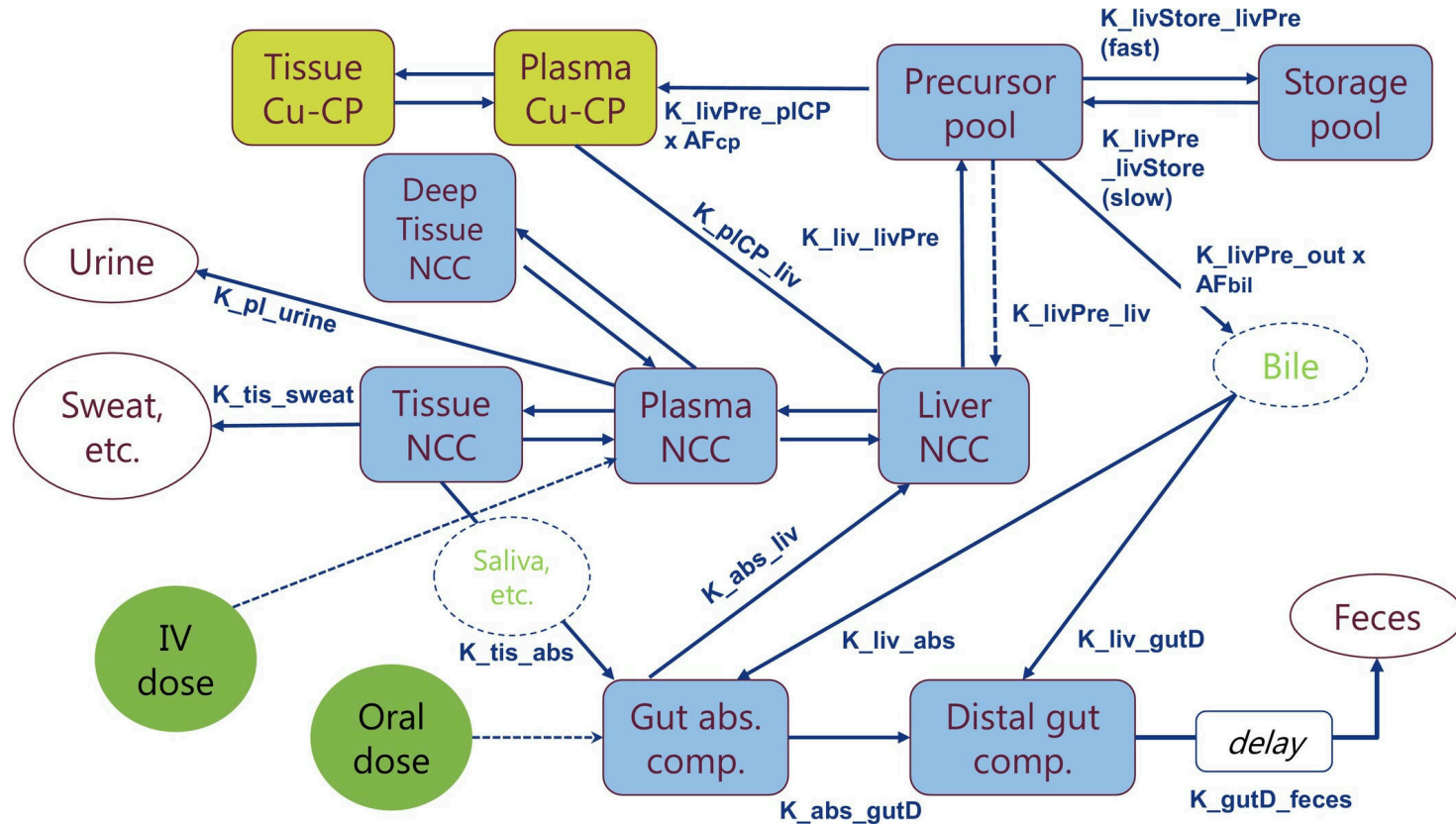


Figure 1: Schematic representation of updated copper model.  $AF_{bil}$ , activity factor for transporter-mediated excretion of copper into the bile;  $AF_{cp}$ , activity factor for bi

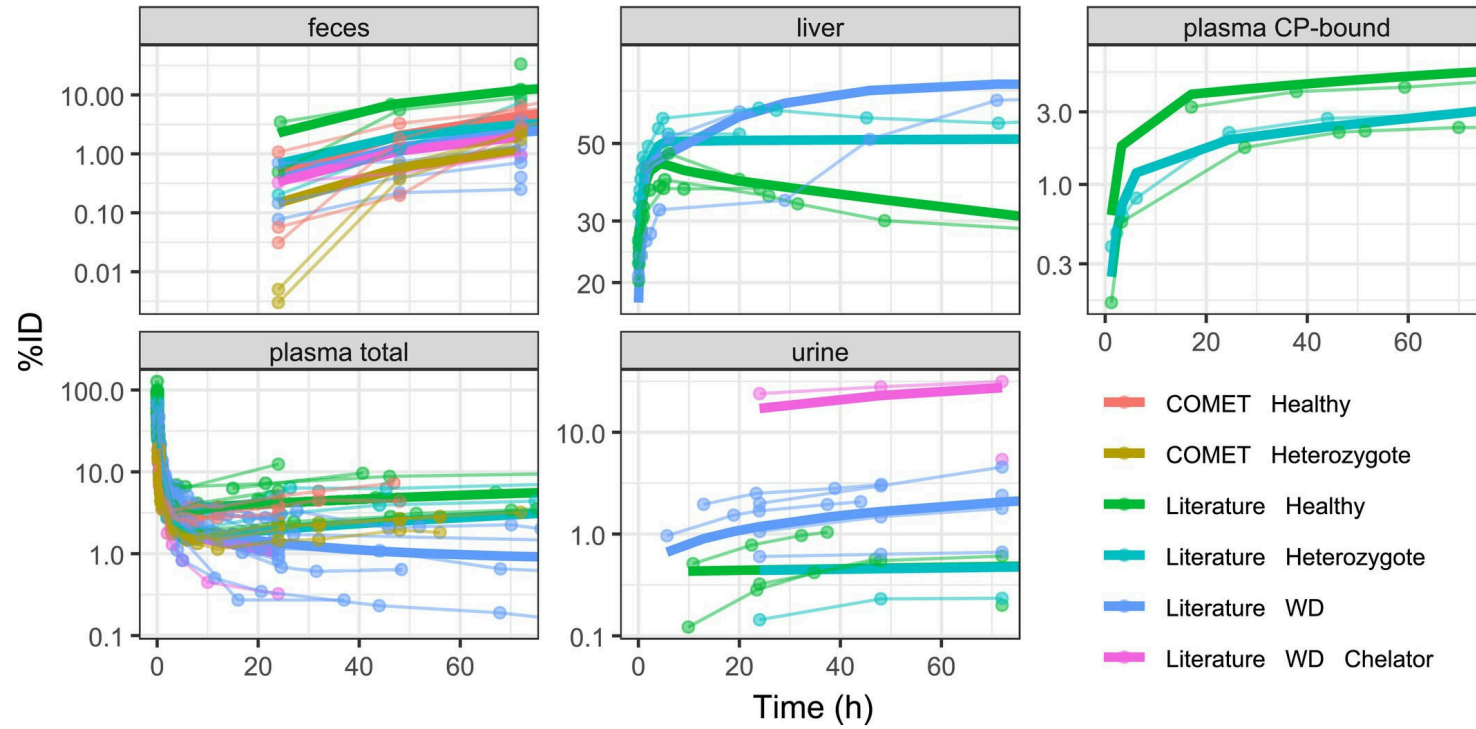


Figure 2: Observed and predicted radioactivity-time profiles in different matrices. Thick solid lines are model predictions, dots and thin lines are observations. %ID, per

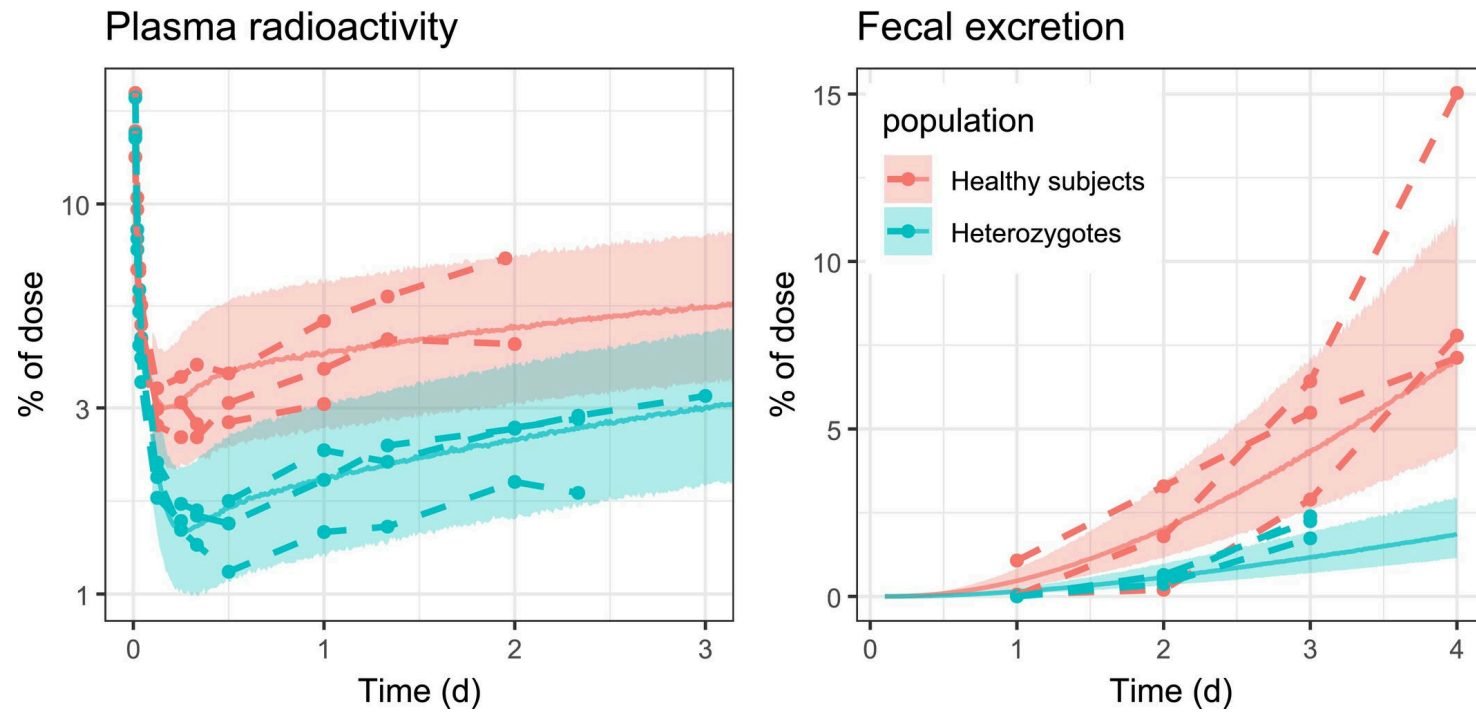


Figure 3: Visual predictive check for COMET data—low variability scenario. Circles are the observations in the COMET study. The shaded areas comprise the 5th–95th percenti

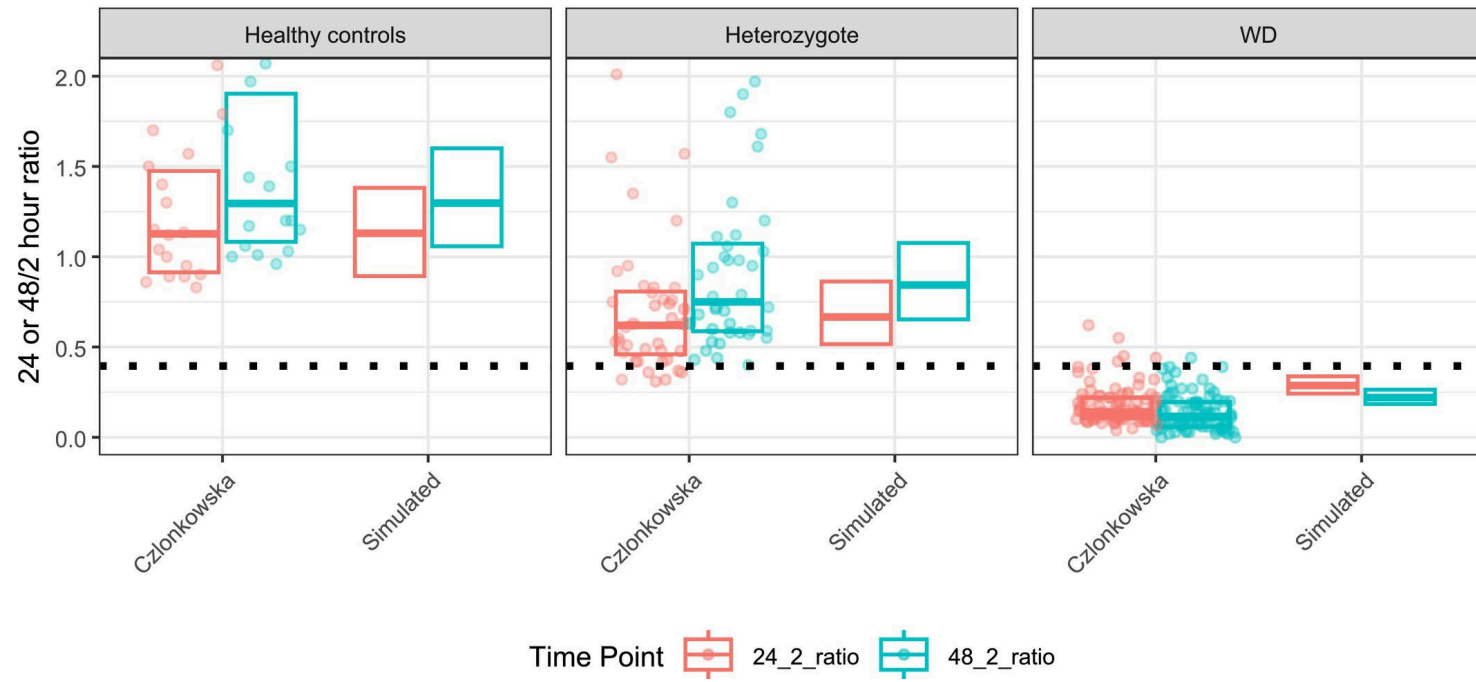


Figure 4: External validation comparing simulated and observed 48/2-h and 24/2-h ratios. Dots are individual measurements, boxes encompass the 25th to 75th percentile of t

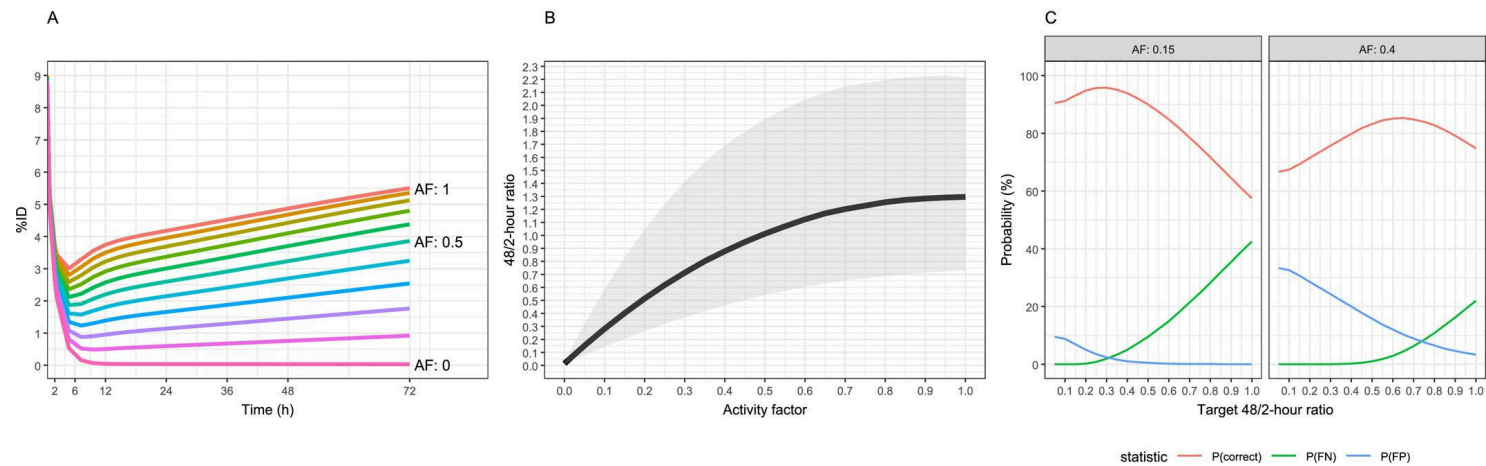


Figure 5: (A) Simulated typical profiles of radiocopper in plasma for different values of the activity factor (AF). (B) Simulated 48/2-h plasma radioactivity ratio as func

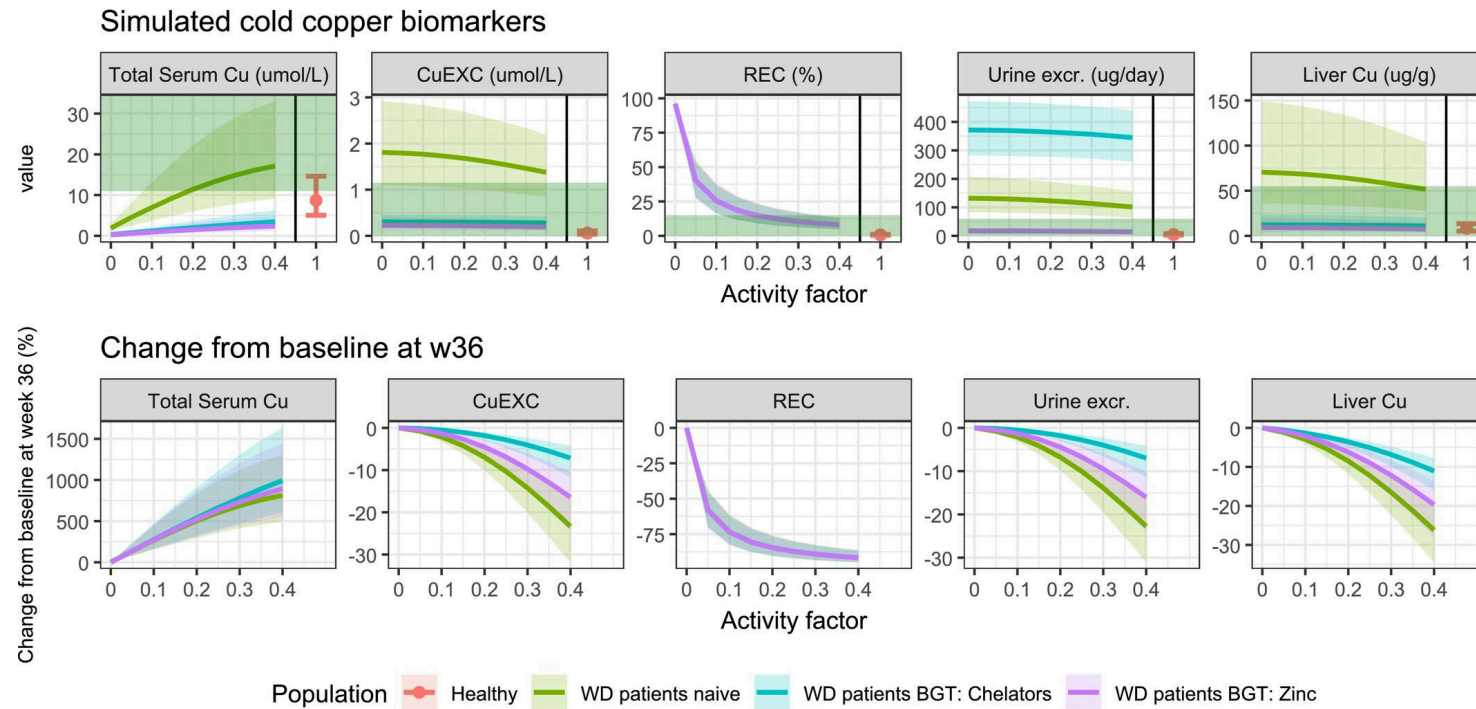


Figure 6: Simulations of cold copper biomarkers. Top row: Raw values. Green areas illustrate areas below or above common diagnostic threshold values (i.e., healthy if with